

PROJECT DESIGN PHASE

SOLUTION ARCHITECTURE DOCUMENT

Import Data Using Transform Maps (Spreadsheet)

ServiceNow Data Import and Transformation

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Team ID	SWTID- 2026- 6497
Project Name	Import Data Using Transform Maps (Spreadsheet)
Technology / Skill	ServiceNow

Prepared for Project Design Phase

1. Project Overview

Project purpose. This micro project focuses on importing structured data from an external spreadsheet into ServiceNow using Import Sets and Transform Maps. The project represents a real-world scenario in which bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately.

Core approach. The spreadsheet data is first loaded into an Import Set table, which acts as a staging area. A Transform Map is then configured to map source fields to the appropriate target table fields. After transformation, records are created in the target table and validated to support data integrity.

2. Problem Statement

Bulk employee information supplied in spreadsheet form needs to be brought into ServiceNow through a controlled import process. The solution must provide a clear flow from spreadsheet data loading through transformation, validation, duplicate prevention, and reporting.

3. Proposed Solution

- Load structured spreadsheet data into a ServiceNow Import Set table.
- Use the Import Set table as the staging area for incoming data.
- Configure a Transform Map to connect source spreadsheet fields with target table fields.
- Execute the transformation so that records are created or updated in the target table.
- Validate the transformed records to check data integrity.
- Use coalesce to help avoid duplicate records during transformation.
- Create reports and dashboards to present the resulting data.

4. Solution Architecture

The architecture bridges the business requirement for reliable bulk employee-data migration with the ServiceNow implementation. The main components and flow are shown below.

Layer	Component	Purpose
Source	Excel / Spreadsheet	Provides structured employee data for import.
Staging	Import Set Table	Receives and temporarily stages incoming spreadsheet records.
Transformation	Transform Map	Maps source fields to the appropriate target fields.
Target	ServiceNow Target Table	Stores the transformed records.

DATA FLOW

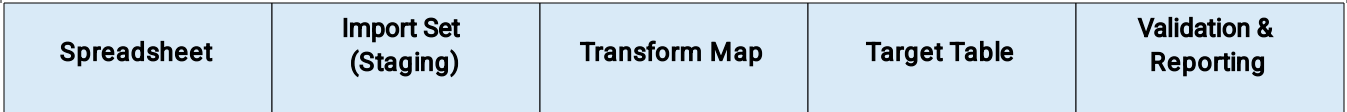


Figure 1: High- level architecture and data flow for the spreadsheet import solution.

5. Architecture Components

Component	Role	Expected Outcome
Spreadsheet	External structured source of employee data.	Prepared source data available for import.
Import Set Table	Staging area for incoming records.	Spreadsheet records are loaded into ServiceNow.
Transform Map	Defines field mapping between source and target.	Source values are transformed into target records.
Coalesce	Used to avoid duplicate records during transformation.	Duplicate creation is reduced.
Validation	Checks the transformed result and data integrity.	Imported records can be reviewed for correctness.
Reports & Dashboards	Presents imported data in a useful format.	Project results can be monitored and reviewed.

6. End- to- End Process

Step	Activity	Description
1	Prepare spreadsheet data	Structured employee data is provided in spreadsheet format.
2	Create/load Import Set	The spreadsheet is loaded into an Import Set table used as the staging area.
3	Create Transform Map	Source fields are mapped to the appropriate target table fields.
4	Transform data	The Transform Map processes the staged records into target records.
5	Validate results	The transformed records are

		checked to support data integrity.
6	Apply coalesce	Coalesce is enabled as required to help prevent duplicate records.
7	Create reports and dashboards	The resulting data is presented through reports and dashboards.

7. Project Milestones

Milestone	Activity	Complexity / Duration	Key Result
Milestone 1	Creation of Spreadsheet and Table	Medium / 40 min	Source spreadsheet and required table prepared.
Milestone 2	Creation of Import Set Table and Transform Map	—	Staging table and field mappings configured.
Milestone 3	Transform Data, Validate and Enable Coalesce to Avoid Duplicate Records	—	Data transformed and duplicate prevention/ validation addressed.
Milestone 4	Creation of Reports & Dashboards	—	Imported data presented for review.

8. Functional Requirements

- The solution shall accept structured data from an external spreadsheet.
- The solution shall load spreadsheet data into a ServiceNow Import Set table.
- The solution shall use a Transform Map to map source fields to target table fields.
- The solution shall transform staged data into records in the target table.
- The solution shall provide a validation step after transformation.
- The solution shall use coalesce to help avoid duplicate records.
- The solution shall provide reports and dashboards for the resulting data.

9. Data Mapping and Transformation

The Transform Map is the central transformation component. Source fields from the spreadsheet are associated with the corresponding fields in the target table. During execution, the staged records are processed according to the configured mappings. The project description specifically identifies field mapping, transformation execution, result verification, and coalesce- based duplicate avoidance as key activities.

10. Validation and Data Integrity

Verify that records loaded into the Import Set table are available for transformation.
Confirm that source fields are mapped to the intended target fields.
Review transformed records in the target table.
Check for duplicate- record scenarios and use coalesce where required.
Verify the final imported data before presenting it through reports and dashboards.

11. Solution Design Goals

Design Goal	Description
Accuracy	Maintain correct mapping between source and target data.
Efficiency	Support bulk migration of spreadsheet-based employee data.
Traceability	Keep a clear flow from source spreadsheet through staging and transformation.
Data Integrity	Validate transformed records and address duplicate- record scenarios.
Usability	Provide understandable reports and dashboards for reviewing results.

12. Project Deliverables

Spreadsheet- based source data prepared for import.
ServiceNow Import Set table used as the staging area.
Configured Transform Map with source- to- target field mapping.
Transformed and validated target records.
Coalesce configuration for duplicate avoidance.
Reports and dashboards for project results.
This Solution Architecture Document.

13. Conclusion

The proposed solution provides a structured ServiceNow workflow for importing spreadsheet data. By separating data loading, staging, transformation, validation, duplicate prevention, and reporting, the architecture provides a clear end-to-end approach for the project. The design follows the project objective of helping beginners understand the ServiceNow data import process, including data loading, field mapping, transformation execution, and result verification.

14. Reference

Solution Architecture template provided for the Project Design Phase. The template

describes solution architecture as the bridge between business problems and technology solutions and identifies its goals as defining structure, characteristics, behavior, features, development phases, requirements, and delivery specifications.